Agentic Institutions: Mandates, Capabilities, Constraints, and Escalation for Machine Economic Actors

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Abstract

Delegated cognition is becoming a dominant mode of knowledge production, yet the institutions governing it remain ad hoc. This paper does three things. First, it states the condition under which a persistent hub model should delegate drafting to worker models rather than draft inline: with inline characters re-paid on every remaining turn T of a session and delegated characters paid once, delegation has lower expected cost when $(w/h) < p(1+T)(1-\alpha)$ up to overhead terms, where w/h is the worker-to-hub price ratio, p the probability a dispatch is accepted, and α the fraction of worker output the hub admits into its context. The condition is the rent-or-buy breakeven with a horizon, stated under an assumption the paper names (inline output persists in context for the session) and shows its own hub violates. For free workers ($w = 0$) delegation wins whenever p exceeds a small threshold set by the cost of a failed attempt, about 0.02 in our runs. Per unit of context budget, delegation admits up to $1/\alpha$ artifacts per inline artifact. Second, it conjectures that as worker cognition approaches zero marginal cost, the binding constraint migrates to hub verification, and it argues that a four-layer institutional anatomy (mandate, capability registry, constraints, escalation) plus a post-hoc provenance audit is consistent with that cost structure, without claiming to derive it. Third, and this is the contribution, it instantiates every measured quantity in a live institution, with the limits of that instantiation stated: one human-supervised hub dispatching to free-tier workers produced this manuscript, and the dispatch script, instrument table, frozen turn archive, incident log, and reviewer transcripts are public. Of the four layers, the mandate and needle-constraint layers ran; the registry and the escalation ladder are specified but did not run. The instrument logged 16 rows (13 briefs, 3 connectivity probes), all of which completed; the needle outcome for those rows is hub-attested, not logged, the dispatcher's own needle test could be satisfied by the CLI's echo of the brief, and the paper's own incident record shows at most 11 of 13 briefs were accepted as delivered on the first attempt. The hub-audited admission fraction ranged over [0.09, 0.19] across four waves. Two incidents, a worker that wrote outside its mandate under a variant of its needle and a fabricated worked example that passed its needle and two rounds of hub-briefed, non-blind agent review, were caught by provenance audit rather than by the escalation ladder. Three further incidents, all hub-side number errors, were caught by the agent reviewers and missed by the audit. That is the paper's practical lesson: needles are not acceptance, consistency-checking review does not stop fiction that fits the frame, and provenance checks must be run by someone other than the author of the number.

1. Introduction

1.1 Machine money

Money is moving between machines. Agent wallets hold balances. The x402 protocol [7] enables per-request payment. Google's A2A protocol [2] standardises how agentic services discover and transact with one another. The infrastructure layer for machine economic actors is in production now.

A fundamental question remains unanswered: who is accountable when an agent acts? Not who built it or who deployed it, but who authorised it, under what constraints, and what happens when it fails.

1.2 The gap

The protocols being standardised address communication, identity, and payment. They do not address authority. An agent may know how to send a message, authenticate itself, and receive payment. It does not know whether it was authorised to initiate the action, what budget it was allocated, or where to escalate when it meets a boundary it was not designed for.

This is the distinction between orchestration and institution. Orchestration coordinates tasks. An institution coordinates authority. It defines what an actor is permitted to do, what capabilities it has demonstrated, what constraints bind it, and what happens when it reaches the edge of its mandate. Human institutions developed these structures because coordination problems at scale demand them, and the formal study of institutions for software agents is a quarter of a century old [31]. The same problems arise when LLM agents organise at scale with real money attached.

1.3 The move

We propose treating agent organisations as institutions. A mandate replaces the job description. A capability registry replaces the résumé. Constraints replace the policy manual. An escalation path replaces the management chain. A post-hoc provenance audit replaces the auditor. Each layer has an economic reading in the cost model of Part 3. The mandate bounds the delegation. The registry enables routing. The constraints make acceptance measurable and enforce budgets. Escalation handles the case the brief did not anticipate. The audit catches what the constraints cannot see.

The anatomy is consistent with the cost model, in the sense that each layer sets one of the variables the model prices. It is not derived from it. §3.5 says exactly what that consistency amounts to.

1.4 The evidence

We operate a live hub-and-worker institution: one orchestrating model manages free-tier worker models that draft artifacts and return results. The economics are measurable. In the novelty-audit wave, the one wave whose produced-byte count reproduces exactly from the dispatch instrument, workers produced 50,567 bytes at zero marginal cost and the hub admitted about 7,000 characters, a hub-audited figure with no instrument record, into its context. Across the four drafting waves the hub-audited admitted fraction ranged from 0.09 to 0.19 (Table 1, §3.1).

Part 3 states the delegation condition and measures its terms. Three things follow from it. For free workers, delegation wins whenever p exceeds a small threshold p* set by the cost of a failed attempt (§3.3). For paid workers, delegation wins if the session outlives a breakeven horizon, under the assumption that inline output persists in context for the session. Our hub's own store compacts, so the measured result is the free-lane one. Per unit of context budget, delegation admits up to $1/\alpha$ artifacts per inline artifact, between 5.3 and 11.1 on hub-audited α, as an upper bound.

The paper also records what the institution got wrong. A worker wrote outside its mandate and signed a variant of its acceptance needle, which the instrument recorded as a completion (INC-001). A worker-generated worked example with no instrument record passed its needle and two rounds of agent review (INC-004). The hub itself destroyed two sections of its own manuscript (INC-002), shipped a stale figure (INC-003), reported a count from memory instead of from a query (INC-005), and carried the query's anchor from a reviewer footnote without verification (INC-006). Every one of these is in Appendix B and in the public incident log, and the controls that caught them are named in §4.4 and §4.6.

Part 2 surveys related work. Part 3 presents the cost model and measurements. Part 4 presents the institutional anatomy with maturity labels. Part 5 describes the method by which the institution studied itself, including what its agent reviewers missed. Part 6 states conjectures for agentic market design.

2. Related Work

2.1 Agent-interoperability protocols

The current stack splits interoperability into two layers. MCP [1] standardises the boundary between a model and its tools: a server exposes capabilities and any compliant client can call them. A2A [2] sits a layer up and standardises agent-to-agent communication: agent cards for discovery, task and message exchange, and capability negotiation across heterogeneous agents. ANP [3] pushes toward a transport-level federated agent network with routing and identity. Together these remove hand-built point-to-point integration. Gap: they standardise how agents talk, discover, and call tools, but are silent on why an agent may act, what authority it holds, what it is forbidden to do, and who answers if it violates a boundary.

2.2 On-chain agent identity and reputation

ERC-8004 [4] provides an on-chain primitive for AI agents: tokenised, verifiable identity whose outputs and provenance are attestable on Ethereum. Adjacent standards extend the identity surface. ERC-4337 [5] account abstraction enables smart-contract wallets operated autonomously by agents, ERC-5192 [6] soulbound tokens give non-transferable reputation anchors, and ENS gives a readable name. NostrAgent [28] proposes a decentralised identity and delegation architecture off-chain. The result is a plausible substrate for machine identities with persistent, globally auditable history. Gap: these give agents identity, provenance, delegation keys, and reputation markers, but nothing encodes a mandate's scope, a capability set, an accountability chain, or escalation rules. Identity without governance.

2.3 Agent simulation platforms

Simulation work established that agent societies exhibit institution-like structure. Generative Agents [8] showed about 25 LLM agents forming schedules, relationships, and social memory. Concordia [9] generalised this into a reusable simulation library. OASIS [10] scaled to a social-media platform run by LLM agents with no hard-coded rules, where norms and trends emerged from interaction. Gap: simulated agents act on synthetic stakes inside a sandbox. Nothing enforces institutions on agents holding real economic value, and no mechanism makes institutions resilient to adversarial or mandate-seeking machine actors.

2.4 Agent memory architectures

MemGPT [11] treats context as an operating system's memory hierarchy, paging between a working window and archival storage. Zep's Graphiti [12] builds a bi-temporal knowledge graph of entities and relations over time. Both give agents durable state beyond a single context window. Gap: memory is descriptive, not normative. Nothing in a knowledge graph confers decision authority, records an agent's commitments, or establishes accountability for what it did with its state.

2.5 Multi-agent frameworks

AutoGen [13], CrewAI, and LangGraph [14] wire model calls into workflows (handoffs, tool loops, graph topologies, role assignments) and manage execution. Orchestration is a prescriptive control flow: one process decides which agent runs when, and cooperative behaviour is assumed. Institutions govern relationships among self-interested actors with competing incentives, define authority and constraints, adjudicate disputes, and hold parties accountable. A graph edge says ”call agent B”. An institution says ”agent B may not exceed its mandate, and here is the escalation path if it tries.” Gap: frameworks optimise pipelines under a benign coordinator and provide no mechanism for authority or accountability among agents that do not share the coordinator's objective.

2.6 Machine payments

ERC-4337 smart-account wallets let agents hold funds and authorise transactions [5], and x402 [7] generalises HTTP request-response into pay-per-call micropayments. On-chain settlement executes without a human in the loop. Sun et al. [30] survey this infrastructure across the agent task lifecycle and find that security and

economic guarantees most often fail where verification meets payment settlement, which is the same seam this paper conjectures to be the binding constraint. What remains aspirational is the governance layer around the money: who sets a machine's spending authority, how constraints are enforced, and what happens on violation. Gap: payment rails let agents transact, but nothing binds spending to a mandate or makes an agent accountable for misallocation.

2.7 Economics and governance of delegation

The nearest neighbours to the cost model are in a separate literature. Anchan and Sen [15] formalise the same inline-versus-delegate trade-off as an attention tax against a handoff tax, with human-subject validation. Our condition is the machine-institution analogue. Theirs prices attention per step, ours amortises it over a session's remaining turns under an explicit persistence assumption (§3.3). Theirs assumes a human attention market, ours a free-lane worker supply. The two should be read together. Hill [16] describes evaluation abundance at organisational scale without a mechanism. Our conjecture in §3.5 is a mechanism-level complement: hub admission αS against a hard context budget K. Kolt [17] and Johnson and Naimi [18] treat governance-by-design for AI agents. Principal-Agent RL [19] treats delegated control. Sreenivas and Larson [20] treat contract structure for LLM delegation. Jensen and Meckling [21] remain the classical anchor, and the premise that monitoring, not production, is the rationed factor in a firm is theirs and Alchian and Demsetz's [33], with Coase [32] behind both. Our first-pass model brackets the moral-hazard channel (p is exogenous. There is no effort choice). The incidents in §4.4 are the empirical evidence that the channel is real.

In the routing literature, FrugalGPT [23] and RouteLLM [24] optimise per-query price among paid tiers. The difference between their condition and ours is one term, the remaining-turns horizon (1+T), and §3.6 shows that term disappears for any hub that can evict its own inline output. Under eviction, at zero overhead, our condition reduces to theirs (§3.6). The cost condition itself has the form of the rent-or-buy breakeven of Karlin et al. [34]. What is ours is the horizon term under a stated assumption and the measured instantiation. Static-context executor architectures [25] extend one agent's window. The per-context-budget multiplier for an institution is unclaimed there.

Formal institutions for software agents predate LLMs. Esteva et al. [31] specify electronic institutions (roles, scenes, norms) for agent societies, and that programme and its successors already bind authority, constraints, and sanctions over agents. The gap this paper addresses is narrower than "no prior institutional structure": it is the absence of such a structure over machine economic actors with live payment rails and LLM workers, grounded in a measured cost model and an instrumented run.

3. The Economics of Delegated Cognition

3.1 Setup and measurements

One orchestrating model (the hub) holds the mandate and the context. Free-tier worker models execute briefs. All quantities below are measured from the drafting sessions of 7–9 October 2026, not simulated. The instrument table, its dispatch script, and every other artifact cited here are public (§5).

Table 1. Delegation waves. "Produced" is bytes of worker output as logged by the dispatch instrument (`wc -c` over the worker transcript, which includes CLI chrome; §3.7). "Admitted" is characters the hub retained in context after verification, a hub-audited figure with no instrument record for any wave, and it can include hub expansion of the worker text, so α is not bounded by 1 by construction. N is dispatches. α = admitted / produced on the hub-audited denominator. Wall is the longest single worker duration in the wave (parallel dispatch), from the instrument.

Wave	N	Produced (bytes)	Admitted (chars)	α	Wall (s)
1. Drafting †	2	22,824 (instr. 2,835)	3,853	0.17	43
2. Intro, citations ‡	2	23,440	4,447	0.19	109
3. Novelty audit	2	50,567	≈7,000	0.14	525
4. Six sections ‡	6	40,447	≈3,500	≈0.09	163

Wave	N	Produced (bytes)	Admitted (chars)	α	Wall (s)
Total, 1–4	12	137,278	18,800	0.137	840 (Σ max)
5. Resolver (not woven)	1	19,759	n/a	n/a	234

† Wave 1 is dual-sourced. The dispatch instrument logged 1,397 bytes for the outline brief and 1,438 for the related-work brief (2,835 first-pass bytes; rows 1 and 2 of the public table). The hub-audited figure of 22,824 counts the artifact at weave time, after hub expansion. The raw worker outputs for 7 October no longer exist because the dispatcher wrote every worker's output to one shared temporary file that later dispatches overwrote (§3.7), so 22,824 cannot be reconstructed and the two figures are shown side by side rather than reconciled. The wave-1 α of 0.17 is hub-audited against the 22,824 denominator; against the instrument denominator the admitted count exceeds the first-pass output, so the ratio is not defined. On rows 1 and 2 the brief file is larger than the logged transcript (1,615 and 1,741 bytes against 1,397 and 1,438), which the earlier instrument version cannot explain and the retained one would not produce. Readers should treat wave 1 as the least reliable row.

‡ Waves 2 and 4 diverge from the instrument by smaller margins (23,440 vs 20,692 and 40,447 vs 40,441) for the same weave-time reason. Wave 3's produced count matches the instrument exactly (30,430 + 20,137). All four instrument sums reproduce from the public table. Row 5 is the citation-resolver dispatch. Its output was used for adjudication and not woven, so it has no admitted count and is excluded from the totals.

Admitted counts. No admitted count for any wave is instrument-logged. The admitted column of the instrument exists from 8 October 2026 but no row carries a value (§3.7). Waves 3 and 4 are reported to the nearest hundred because the hub did not record the exact figure. All four α values are therefore hub-audited estimates, and the measured results that depend on them (§3.4, Appendix A F1) are labelled as bounds.

Units. Costs are in characters of hub context, and the instrument logs bytes. For English prose the two differ by a few percent plus CLI chrome, and w/h as a per-character price ratio absorbs the difference only if numerator and denominator are measured on the same text. We say "bytes" wherever the instrument is the source.

Pricing assumptions. Assumption P1: hub input and output characters are priced uniformly at h, worker characters at w. Assumption P2: every re-read of a persisted hub character costs h. Vendors that discount cached-prefix input charge c·h for a re-read with $c < 1$; under caching the horizon term (1+T) below becomes (1+cT) and every breakeven horizon in §3.3 scales by 1/c. Whether the gateway used here applies such a discount is not established; §3.3 gives one sensitivity row.

3.2 The two cost laws

Hub attention is persistent: a character in the hub's context is re-paid as input on every remaining turn of the session. A turn here is one API invocation of the hub model. If the session has T further turns after the artifact exists:

Inline drafting (the hub writes it): $C_in = h \cdot S \cdot (1 + T)$. Write once, re-read every turn.

Delegation (workers write, hub verifies): $C_del = (w/p) \cdot S + h \cdot (\alpha S + b + o) \cdot (1 + T) + h \cdot \varepsilon \cdot (1/p - 1) \cdot (1 + T)$.

Here p is the single-attempt probability that a dispatch is accepted as delivered, b is the characters of brief the hub writes, o is the characters of dispatch tool output that land in hub context, and ε is the hub characters consumed per failed attempt (failure verdict plus re-brief). The (w/p)S term is the expected worker cost under retry-until-accept (geometric attempts, mean 1/p). Everything the hub writes or reads persists under the same law as the artifact, which is why b, o, and ε carry the (1+T) factor. Our instrument logs none of o or ε. b is recoverable from the committed briefs: the twelve drafting briefs in `briefs/` total 19,262 bytes against 114,535 instrument-logged output bytes, $b/S = 0.17$ pooled over the twelve on the instrument denominator (0.14 with rows 1–2 excluded, or with b and α both on the hub-audited denominator of Table 1), 0.09–0.38 on the ten briefs whose logged transcript exceeds the brief and above 1 on rows 1–2, with two caveats: the committed copies are post-typography-pass (bytes within 1 %), and on rows 1–2 the brief exceeds the logged transcript

(1,615 vs 1,397 and 1,741 vs 1,438 bytes), which is noted beside Table 1 note †. §3.3 states what dropping the terms costs.

Lemma 1 (retry cap). If the hub caps attempts at R and drafts inline on exhaustion, with q = 1 − p, then C_del^R = (1 − q^R)·C_del + q^R·[C_in + h(b + o)(1 + T)], because the brief and dispatch output are paid once per dispatch whether or not it ends in acceptance. Hence C_del^R < C_in iff C_del + (q^R/(1 − q^R))·h(b + o)(1 + T) < C_in. The residue vanishes at b = o = 0 and is below 0.4 % of the brief term at p = 0.85 and R = 3 (but 52 % at p = 0.3, the grid's lowest row), so the condition below is the same for bounded and unbounded retry up to that residue. The lemma assumes the inline fallback costs exactly C_in and that each failed attempt leaves only ε in context. It reconciles this model with a retry cap of the kind the ladder of §4.5 specifies. The ladder's terminal rung, a human reviewer, is not priced by the model, and the residue is negligible at the measured p, not at the grid's lowest row.

3.3 The delegation condition

Proposition 1. Under the cost laws of §3.2 and Assumptions P1–P2, delegation has lower expected cost than inline drafting iff

```
(w/h) < (1 + T) · [ p · (1 − α − (b + o)/S ) − ε · (1 − p) / S ].        (⋆ε)
```

Proof: divide C_del < C_in by h·S·(1+T), collect terms, multiply by p(1+T).

Dropping the overhead terms (b + o)/S and ε(1−p)/S gives the simplified form used for the sensitivity grid below:

```
(w/h) < p · (1 + T) · (1 − α).        (⋆)
```

(⋆) overstates delegation's advantage by exactly the dropped terms, so every horizon derived from it is a lower bound on the true breakeven horizon. In our runs failure tails were hundreds of characters against artifacts of tens of thousands, so ε(1−p)/S is below 0.02, and b/S = 0.17 pooled over the twelve drafting briefs (§3.2), while o remains unlogged.

Assumption E (persistence). (⋆ε) charges inline output on every remaining turn. That requires inline output to stay in hub context for the session. §3.6 shows this hub's store compacts, which violates E, and states what survives without it.

Corollary 1 (free lane). For w = 0 and (b+o)/S < 1 − α, (⋆ε) holds iff p > p* := ε / (ε + S(1 − α − (b+o)/S)). At the magnitudes of our runs ($\varepsilon \approx 300$, S ≈ 20,000, $\alpha = 0.17$, o unlogged and set to zero) p* ≈ 0.018 with b set to zero and 0.022 at the measured b/S = 0.17 on the instrument denominator (0.021 at b/S = 0.14 on the hub-audited denominator), which we round to 0.02 throughout. Free-lane delegation therefore wins for any dispatch accepted more than about one time in fifty, and that is the only form in which this paper claims it.

Proposition 1, in the form (⋆), is the rent-or-buy breakeven of Karlin et al. [34] with horizon 1+T: the one-time cost of delegating, (w/p)S, against the per-turn rent the hub pays to keep inline output in context, h(1−α)S. The paper claims no novelty for the form. The classical online strategy for an unknown horizon (rent until accumulated rent equals the buy price, then buy) does not transfer directly, because switching from inline to delegated mid-session would require evicting the inline draft, which Assumption E says the hub cannot do. In the no-switch regime the two committed policies have simple bounds against the T-informed optimum: inline-always costs at most $1/\alpha$ times the optimum (about 7 at the pooled α), and delegate-always at most max{1, (w/h)/p + α} times it. For free workers (w/h)/p + α = α < 1, so delegate-always is the optimum itself (ratio 1) with no knowledge of T required. With overhead restored its cost relative to inline is α + (b+o)/S + ε(1−p)/(pS), below 1 exactly when p > p* (Corollary 1). The hub is taken to be risk-neutral throughout: the retry-until-accept cost has unbounded support and only its expectation is compared.

Paid workers: sensitivity grid. The ratios w/h ∈ {15, 20, 30} are an adversarial range, not a realistic regime: in practice workers are cheaper than the hub, not dearer. Breakeven horizon T from (⋆) at $\alpha = 0.17$ (wave 1):

p	w/h = 15	w/h = 20	w/h = 30
0.3	$T > 59.2$ ($T \geq 60$)	$T > 79.3$ ($T \geq 80$)	$T > 119.5$ ($T \geq 120$)
0.5	$T > 35.1$ ($T \geq 36$)	$T > 47.2$ ($T \geq 48$)	$T > 71.3$ ($T \geq 72$)
0.7	$T > 24.8$ ($T \geq 25$)	$T > 33.4$ ($T \geq 34$)	$T > 50.6$ ($T \geq 51$)

Sensitivity. At the pooled $\alpha = 0.137$ every horizon falls by about 4 % (p = 0.3: $T > 56.9$ / 76.2 / 114.9). With the measured brief term $b/S = 0.17$ restored, every breakeven $1 + T$ rises by a factor $(1 - \alpha)/(1 - \alpha - b/S) \approx 1.25$ (p = 0.3: $T > 74.5$ / 99.7 / 150.1), and the largest cell, $T > 150.1$, is still cleared by the 221 distinct turns of the session below. At a cached re-read price $c = 0.1$ (Assumption P2) every horizon rises tenfold, and the p = 0.3, w/h = 30 cell becomes $T > 1{,}195$, which the session below does not clear. The mandate-compliant acceptance rate on the paper's own record is at most $11/13 \approx 0.85$ (§3.7), above every row of the grid. All of this is conditional on Assumption E.

The remaining-turns count. Working sessions in our logs are long. The hub's message store was queried for assistant rows in the hub session after the drafting wave:

```
sqlite3 state.db "SELECT COUNT(*) FROM messages
  WHERE session_id='20260725_131848_6f3bc9c8' AND role='assistant'
  AND timestamp >= strftime('%s','2026-10-07 13:18:00');"
```

The anchor string is interpreted by SQLite as UTC. The event it names is the INC-001 dispatch, appended to the instrument at 13:18:34 local time (UTC+4), 09:18 UTC. The other five section dispatches were appended at 13:16. Because every row from 7 October had been evicted by the time of the freeze, both anchors select the same set. The query returned 343 rows at the freeze, commit cd46a7e of the companion repository, authored 2026-10-09 04:16:35 +04 (00:16 UTC). The qualifying row ids and timestamps are archived in the repository as `data/t_freeze_2026-10-09T0416.csv`, SHA-256 5572cf45f6c551739a4e7ed5afabebaf45f40deae28c27b85f0eafcd55eb4258, and `scripts/verify_t.sh` recounts the archive and checks the hash. The archive is a frozen receipt that the count was taken. It is not an independently checkable measurement, because the store it was taken from is not public.

Raw rows are not turns. Of the 343 archived rows, 122 share one of four second-resolution timestamps (48, 18, 26, and 30 rows at 2026-10-08 08:46:41, 14:13:40, 15:21:55, and 17:42:43 UTC) with ids stepping by two inside each burst (two gaps of three in the last burst). No model emits 48 turns in one second. We infer that these are compaction rewrites of earlier turns. The archive carries only id and timestamp, so the inference rests on the spacing, and the store columns that would confirm it (role, active flag, finish reason) are not public. There is no timestamp shared by two, three, or four rows, so the threshold of five does no quiet work. If a future freeze's de-duplicated count fell below the largest horizon in the grid, the paid-lane sensitivity claim would be withdrawn for that session. Excluding every burst row leaves 221 distinct assistant turns, and `scripts/dedup_t.sh` reproduces that count from the archive. Both figures clear the largest horizon in the grid ($T > 119.5$). An earlier freeze on 8 October returned 277 raw rows, and a re-run on a machine without the hub's store returns 0. Because the store compacts, a turn that was evicted at turn 50 of 221 was re-paid 50 times, not 221, so even the de-duplicated count is an upper bound on the horizon term ($\star\varepsilon$) prices, not a floor.

3.4 Observation: yield per unit of context

A hub session has a context budget K, in characters. Inline, each artifact consumes S of it. Delegated, each consumes αS plus the brief, the dispatch output, and any failure residue. Writing $\Delta = \alpha S + b + o + \varepsilon(1/p - 1)$ for the per-dispatch context delta:

```
N_in = K/S,   N_del = K/Δ,   N_del / N_in = S/Δ ≤ 1/α.
```

This is an identity, not a result: $1/\alpha$ is α's definition inverted. Its content is the bound. On hub-audited α the admission-only bound $1/\alpha$ is between 5.3 and 11.1 across waves (wave-rounded 0.19 and 0.09) and 7.3 pooled. With the measured brief term, $S/\Delta \leq 1/(\alpha + b/S) = 3.3$ pooled, with α hub-audited and b/S on the instrument denominator (3.6 with both on the hub-audited denominator), and 3.0 at the wave-1 α with the pooled brief term (3.2 on wave 1's own hub-audited denominator), and the true multiplier is below that by the still-unlogged o and ε terms. We do not call this a lifespan multiplier: the hub session in Appendix A F3

has survived multiple compactions rather than dying when K filled, so "lifespan" would need an operational definition (turns until first forced compaction) that we have not measured. What the bound says is that per unit of context, a hub that admits a fraction α of worker output can process up to $1/\alpha$ times as many artifacts as one that drafts inline, and that the hub's admission policy, not delegation as such, is what sets α (§3.6).

3.5 Conjecture: verification binds, and the anatomy is consistent with the cost model

Jevons [22] observed that efficiency in coal use increased total coal consumption. Narayanan and Pace [26] ask whether the same rebound applies to AI labour markets. We conjecture that it governs cognition inside a single institution: as the marginal price of worker cognition falls toward zero, dispatch volume expands until the hub's context budget binds. The two-constraint sketch is: a hub choosing dispatch volume D to maximise value V·D faces $D \cdot \Delta \leq K$ (context) and $D \cdot (w/p) \cdot S \leq B$ (money). As $w \to 0$ the money constraint's shadow price goes to zero and the context constraint binds. Nothing in this paper measures D expanding or K saturating, and the cost model charges the hub nothing for verification itself beyond the admitted characters, so the paper cannot show that verification binds. This is a conjecture, labelled as such wherever it appears, and [16] leaves the organisational-scale version open.

What ($\star\varepsilon$) does support is a consistency claim. Its variables are set by identifiable institutional choices. The brief fixes S and what counts as done (mandate). The choice of worker fixes w and p (capability routing, which in the single-worker case reduces to the cascade rule of [23][24]: pick the worker minimising w/(hp)). The acceptance criterion fixes how p is measured and what ε costs (constraints). The retry and rewrite policy fixes ε (escalation). The post-hoc audit fixes whether a needle hit is counted as acceptance at all (provenance audit, §4.4). We call the anatomy consistent with the cost model in that sense and no stronger one. We do not claim the layers are derived from ($\star\varepsilon$) or that they are the unique policy that satisfies it. Any design that leaves w, h, p, T, and α unchanged is equally consistent with it. The claim that machine coordination needs institutions because monitoring, not production, is the rationed factor is the premise of Alchian and Demsetz [33] and Jensen and Meckling [21]. What this paper adds is a per-character, per-turn form of that premise and a measurement of its terms.

3.6 Threats to validity

Eviction (Assumption E fails). C_in charges every inline character on every remaining turn. But the hub's message store compacts: superseded turns are evicted, which is why the T count in §3.3 is an upper bound on presence-in-context. A hub that drafts inline, writes the draft to disk, and lets the drafting turn compact pays approximately $h_out \cdot S + h \cdot \alpha S \cdot (1+T)$, with $h_out = h$ under P1: the same persistence term as delegation, plus one output charge. Take eviction to apply to all hub output alike (the inline draft, the brief, the dispatch output, and the failure residue). Against that baseline the paid-lane condition reduces to $w/h < p - [p(b + o) + \varepsilon(1 - p)]/S$, which at zero overhead is $w/h < p$, the per-query cascade condition of [23][24], and the (1+T) amplification in ($\star\varepsilon$) disappears. The $1/\alpha$ bound of §3.4 is then a property of admission policy, not of delegation: any hub that writes to disk and admits a fraction α of its own output obtains it. Delegation is one way to obtain $\alpha < 1$. In our runs it was the only way available, because the hub cannot evict within a turn and its own drafting output stays in context until the next compaction, which the hub does not control. But the archive in §3.3 shows that compaction did occur, so E held only between compactions. The free-lane corollary survives with its threshold changed: under eviction, free delegation wins iff $p > \varepsilon/(\varepsilon + S - (b + o))$, about 0.015 at our magnitudes with $b = 0$ (0.018 at the measured b), slightly below p* because both sides now pay the admission term. The paid-lane grid should be read as an upper bound on delegation's advantage, valid while inline output persists. Measuring how long inline output persists before compaction is the next instrument addition.

Caching (Assumption P2). See §3.1 and the sensitivity row in §3.3.

T as expected remaining turns. Session length is heavy-tailed and sessions die early, which hurts the paid-lane case because the inline alternative then looks cheaper. The committed-policy bounds in §3.3 are the hedge: for free workers above p*, delegate-always needs no estimate of T.

p. The instrument records completion, not acceptance (§3.7). The rate used in the sensitivity discussion,

$\leq 11/13$, is the paper's own incident-corrected figure, and briefs were self-selected and heterogeneous, so no single p describes them.

Benign workers. All workers were cooperative free-tier models with no incentive to game the needle. The anatomy has not been exercised against an adversarial or mandate-seeking worker, which §2.3 names as the gap in prior work. Its behaviour against such a worker is uncharacterised.

ε **and overhead.** ε was small in our runs (failure tails of hundreds of characters) but is not logged. The brief term b is recovered from the briefs (§3.2) and o is not logged. $p^* \approx 0.02$ depends on a failed attempt leaving only a verdict in context ($\varepsilon \ll \alpha S$). If each failed attempt left its admitted share, the free-lane threshold would be $p > \alpha$ (at zero overhead). All three enter ($\star\varepsilon$) in the direction of making delegation look better than it is.

Wave-1 denominator. The instrument and the hub disagree by 8× on wave 1 (Table 1, note †). Nothing in Parts 3–4 depends on wave 1 alone: the free-lane corollary needs only $p > p^*$, the $1/\alpha$ bound is set by waves 2 and 4, and the grid is given at both the wave-1 α and the pooled α.

Admitted counts. No admitted count is instrument-logged (§3.1). Every α in this paper is a hub-audited estimate.

The free-lane corollary survives all of these. It needs $w = 0$, $(b+o)/S < 1 - \alpha$, and p above a threshold of about 0.02.

3.7 Instrumentation

The dispatch script (`instrument/hf_dispatch.sh` in the companion repository) runs one worker under a wall-clock alarm and appends one row per attempt to an instrument table: date, worker, brief, budget, duration, output bytes, message count, verdict, and (from 8 October 2026) artifact bytes and admitted characters. The frozen snapshot used throughout this paper is the first 16 rows: 13 briefs and three connectivity probes.

What the verdict column records. In the version of the script that wrote rows 7–26 (rows 1–6 came from an earlier version not retained, and on rows 1–2 the logged transcript is smaller than the brief, so the echo mechanism below is established for rows 7–26 only), the row is written before the acceptance grep runs, and the verdict is OK if the worker process exited zero and TIMEOUT otherwise. The needle outcome is not logged. So the frozen snapshot shows 16 of 16 rows completed with exit status zero, and nothing more. The dispatcher's needle test attested nothing about the worker either: the worker CLI's single-query path echoes the full brief to standard output before the response, every brief spelled out its needle in a trailer instruction (verified: all 23 v1 dispatched briefs in the repository's `briefs/` folder end with one, as does the reconstructed v2 probe brief), and the v1 test searched the whole transcript, so on rows 7–26 the test could be satisfied by the echo alone. Needle status for the frozen rows is therefore hub-attested only. Row 13, the six-section wave's last dispatch, is the INC-001 dispatch, whose worker signed a variant of its needle rather than the specified string, whose row reads OK, and whose variant the hub accepted as a signature at the time. On the hub's attestation, 12 of 13 briefs emitted their specified needle.

What acceptance means. A needle hit is not acceptance. Two briefs in the frozen set were counted as accepted at the time and were not: INC-001 (a scope violation under a variant needle) and INC-004 (fabricated content under a correct needle). The mandate-compliant first-try rate on the paper's own record is therefore at most $11/13 \approx 0.85$, and that is the p the sensitivity discussion in §3.3 uses. The exit-zero rate 16/16 carries a Clopper–Pearson 95 % interval of [0.79, 1.00] under an i.i.d. assumption that the self-selected, heterogeneous briefs do not satisfy. The interval bounds a pooled rate over this brief mix, not a property of any brief.

Instrument versions. Rows 1–6 carry minute-resolution timestamps and rows 7–26 second-resolution, so rows 1–6 were written by an earlier version of the script that is not in the repository. Worker outputs are named by needle rather than by brief filename, and the section briefs' filenames do not track their section labels. `briefs/README.md` maps every instrument row to its brief, section, needle, and output file, and should be used instead of the filenames. Timestamps are host-local (UTC+4) without a zone marker. The `admitted_s` column was added on 8 October and is read from an environment variable at dispatch time, before any adjudication exists, so it is empty in every row. The raw worker output was written to one shared temporary path that each dispatch overwrote, which is why no 7 October output survives (Table 1, note †). Each of these is addressed in the v2 script that accompanies this revision:

- The worker is invoked on the CLI's quiet path, which prints the final response alone to standard output (no brief echo, no box chrome, no exit summary) and the session id to standard error, so there is nothing to cut.
- Line endings are normalised before the gate, because the CLI's interactive path writes carriage returns when its output is a file and a hand-written fixture had missed that.
- The row is written after the gate with a four-valued verdict (OK, NEEDLE-MISS, TIMEOUT, ERROR) and the numeric exit code.
- The needle is required as the last non-empty line of the response and its position logged.
- Timestamps are UTC.
- Per-dispatch output and error files are retained (for every row but the removed first attempt, below).
- The append is serialised with a portable lock and a row that cannot take the lock is kept in a pending file rather than lost.
- And the brief bytes (b) and the bytes the dispatcher returns into hub context (o) are logged.

The quiet path does not report a message count, so v2 rows carry none, and o counts the dispatcher's printed summary, which can exceed the raw response (512 against 449 bytes on the one live row). v2 still enforces scope by instruction, not by the dispatcher. v2 has been tested against fixtures generated with the same print call the quiet path uses, including a carriage-return case, and was exercised once against a live worker on 14 October 2026: a probe brief dispatched to DeepSeek returned in 9 s with a two-line response ending in the needle, verdict OK, session id captured (row 1 of `instrument/hf_stats_v2.tsv`, raw output retained). Its first attempt had failed before any worker ran, because the script passed the brief to the wrong CLI switch, and that error row was removed as a script measurement rather than a worker one. No number in this paper comes from the v2 table. The 26 historical rows are left as they are, under the hash recorded in the number snapshot.

Live instrument. The public table holds 26 rows as of 9 October 2026: 24 exit-zero completions and two TIMEOUT verdicts, both on the same round-4 reviewer brief, which exceeded its wall-clock budget twice (17,507 and 21,035 bytes of transcript captured. The v1 artifact column recorded 10,497 and 14,302 bytes after the last occurrence of the needle string, which on those rows was the CLI's echo of the brief). Those are budget exhaustions, not acceptance failures, and the second was a manual hub re-dispatch, not the ladder of §4.5. The second attempt's review was taken from a file the worker had written and was used in round 4. The round-4 brief (`briefs/review4_C.md`) permitted exactly that one file write and forbade every other, so the write was within mandate. The two probe re-dispatches on 8 October (rows 15 and 16) were connectivity checks, not retries. Over all 26 rows the exit-zero rate is 24/26.

Duration at the frozen snapshot ranged 7–525 s with median 29 s over 16 rows (the earlier n = 14 snapshot's median was 31.5 s). Message counts ranged 2–70, because research sweeps make heavy internal tool use.

Caveats. n is small and briefs were self-selected. This is an accumulating instrument, not a result. The traffic asymmetries quoted in Appendix A F3 come from the hub's private message store and use different denominator windows (hub lifetime since 25 July 2026 versus worker artifacts from the 7 October measurement day), so they are not reproducible from the repository and the artifact-only ratio of 185× is cited only as a frozen value.

4. The Anatomy of an Agentic Institution

Sections 4.1–4.5 were drafted by free-lane workers, needle-gated, provenance-audited, and assembled by the hub (§5). Maturity labels: IMPLEMENTED means the mechanism ran in the sessions reported here and has a record in the public artifacts. DESIGNED means it is specified but did not run. Enforcement in the reported sessions: the only constraint enforced by the dispatcher was the wall-clock kill. No-fabrication of citations was instructed in every drafting brief. Scope and single-file writes were instructed only in the nine review briefs (ten dispatches) after INC-001, the only review briefs retained. The 13 drafting briefs, which produced every number in this paper, carried no action constraint at all (`briefs/`), and the dispatcher enforced none in either case. The dispatcher contains no write allow-list, directory jail, or tool restriction. The worker CLI did not restrict file writes (INC-001) or shell and store access (review rounds 3–5), and whether it restricts anything else is not established. INC-001 is what unstated scope looks like when it fails. The ten instructed dispatches after it record no write to the paper tree outside the permitted file (n = 10; detection was the hub's post-wave

scan for round 3, which ran before the repository's root commit, and git for rounds 4–5; the raw transcripts are not retained). Three round-5 workers copied the repository to /tmp to tamper-test the verifier, a write the briefs' literal text did not permit and the hub's tree diff could not see, which each reported and the hub did not treat as an incident. Instruction-only scope held by its intent as the hub read it, not by its letter, and the reading is not logged.

4.1 Institutions as coordination technology (framing, no maturity label)

Human coordination fails without architecture. We do not spontaneously resolve competing claims on shared resources, verify that work was performed, or hold parties accountable when promises break. Firms, contracts, and markets exist because coordination under bounded trust is expensive, and someone has to pay the overhead [32][33]. A firm is an authority-allocation device: it replaces a market for every task it internalises. A contract is a verification and accountability architecture: it specifies what counts as performance, who judges it, and what happens when judgment fails. A market is a dispute-escalation interface: prices aggregate heterogeneous preferences into one comparable signal.

Each of these solves a coordination problem that does not disappear when the actors change. We are now fielding agents that accept mandates, produce artifacts, and hand results to downstream systems. Their coordination problems are structurally the same, and the analogy is the motivation for the layers below, not their justification. §3.5 gives the economic reading.

Authority allocation. In a firm, a principal delegates to an agent and accepts that the agent's judgment may diverge within defined bounds. In a hub-and-worker system, the hub dispatches a brief specifying the outcome but not the method. The brief is a mandate. The worker's authority to choose how is bounded by the acceptance needle in the brief: the task is complete when the output clears the needle and the audit, not when the worker decides it is done.

Verification. Contracts require observable performance. In agent systems, verification takes the form of acceptance tests embedded in the dispatch protocol. A brief defines ”done” as a set of measurable conditions. The record of needles passed or failed would be the agent-era analogue of a countersignature. In the reported sessions that record was hub-attested rather than logged (§3.7).

Accountability. When a contracted party fails to perform, the contract provides escalation: renegotiation, penalty, or exit. In a dispatching hub, accountability would be expressed through acceptance rates and retry routing: a brief that failed its needle would be escalated to a different worker and the failure recorded against the originating worker's registry entry. No registry exists in the reported sessions (§4.3) and no escalation occurred (§4.6).

Dispute escalation. Markets resolve conflicting claims through price. Agent systems would resolve them through escalation chains: a brief that cannot be satisfied within the defined needle space is flagged, rerouted, or decomposed into sub-briefs.

We do not choose whether agents get institutions. We choose whether to design the coordination layer or let it emerge from the accumulated accidents of systems never designed for each other.

4.2 The mandate layer (IMPLEMENTED by instruction, see element labels)

A mandate is a formal delegation contract: a scope, a deliverable, a set of acceptance criteria, and a budget, all made legible before any cognitive work begins. Without this object the principal has no legible stake and the agent has no bounded problem.

The formal object. Let A be the worker's action alphabet (file writes, tool calls, messages) and X the artifact space. A mandate is the tuple $M = (\sigma, \delta, \nu, \beta, \rho)$ with

- $\sigma \subseteq A^* \times X$, the scope: the permitted execution traces and the outputs they may produce (not stated in any drafting brief, where the deliverable implied it; stated by instruction in the review briefs after INC-001; never enforced);
- $\delta \subseteq X$, the deliverable: the set of acceptable artifacts (IMPLEMENTED by instruction);

- $\nu : X \to \{0, 1\}$, the acceptance criterion, computable within the verification share β_ν of the budget β (IMPLEMENTED as a needle grep);
- $\beta \in \mathbb{R}^3$, the budget in tokens, seconds, and cost (IMPLEMENTED for seconds only; token and cost caps DESIGNED);
- $\rho : X_partial \to A^*$, the rollback constraint, a map from a partial outcome (X_partial, the prefixes of artifacts in X) to a recovery action sequence (DESIGNED; no rollback path exists in the dispatcher).

M is well-formed when three conditions hold. (i) ν is sound for δ: every x with $\nu(x) = 1$ lies in δ. (ii) Every execution that produces an $x \in \delta$ and is reachable by the worker lies in σ. (iii) The cost of computing ν is within β_ν. These conditions do work. INC-001 is a trace (t, x) with $x \in \delta$ (the section text was accepted at assembly) and $(t, x) \notin \sigma$: the worker's action, writing into the manuscript, was outside scope, so (ii) failed. Its output also carried a variant of the specified needle, so $\nu(x) = 0$ for the specified ν, which the instrument did not record and the hub did not act on. INC-004 is an x with $\nu(x) = 1$ and $x \notin \delta$: the artifact satisfied the needle but a fabricated worked example is not an acceptable artifact, so (i) failed. A needle sound for the deliverable would have rejected INC-004. A scope enforced over traces rather than over outputs would have prevented INC-001. The provenance audit of §4.4 is the mechanism that checked (i) and (ii) after the fact in both cases. Two consequences follow for the mandate as implemented. A literal needle can never satisfy (i) for any non-trivial δ, so no mandate in the reported sessions was well-formed. And (ii) quantifies over executions reachable by the worker, which an instruction-only scope does not bound, so (ii) could only ever be audited after the fact, never evaluated in advance.

Principal-agent theory [21] identifies the core failure mode: the agent optimises a proxy of the principal's objective because the objective was never specified precisely. In software this appears as agent drift, the agent reinterpreting its task in directions that are locally coherent but globally misaligned. A second failure mode compounds the first: unbounded retries. Without a termination condition, an agent that fails an imprecise deliverable retries until something looks right, with no convergence guarantee. The dispatch protocol bounds this by encoding ν as a hard gate: the brief is not complete until $\nu(x) = 1$, and ν is itself a metered call. The A2A task object [2] encodes this structure natively: a task carries status, output, and a signals block through which the principal injects acceptance signals without owning the execution loop. A mandate is the schema before that instance.

Running example (illustrative, not instrument data). The hub receives "write the introduction for Part 3," decomposes it, and issues a mandate:

- Scope: Part 3 introduction, about 500 words, position-paper voice; write to the returned message only.
- Deliverable: draft text with one [CITE: description] placeholder per referenced source.
- Acceptance (machine-checked): word count in [450, 550]; ends with the literal needle string. Acceptance (human-checked at audit): no fabricated citations.
- Budget: 200k input tokens, 8k output tokens, 120 s wall clock (only the last is enforced by the dispatcher).
- Rollback: on budget exhaustion, return partial output with an [INCOMPLETE] marker.

With this contract the hub's machine check runs in time linear in the artifact and independent of the worker's internal effort, which matters when the hub orchestrates many concurrent workers and its own verification bandwidth is the scarce factor.

4.3 The capability registry (DESIGNED)

Any institution that delegates cognition must answer: who may do what, who can do what, at what price, and how well? Those are two different questions. Capability as granted permission (what the worker may do) belongs to σ in the mandate. Capability as attested skill (what the worker has shown it can do) is what the registry stores. INC-001 was a failure of the first kind: the worker had the permission to write files and should not have. The registry addresses the second kind.

Definition. A capability registry is a dynamic, queryable record of worker abilities, expressed as a vector of task type × quality × cost. It is the institutional analogue of a résumé, but a live attestation of what a worker has demonstrated rather than a static credential. Static configurations (allow-lists, hard-coded roles, fixed tiers set at provisioning) assume capability is a property of identity. It is not. A worker's effective capability

set changes with its model weights, context length, tool access, and load. A registry frozen at deployment becomes noise within weeks.

Probe-based discovery. Rather than pre-declaring that worker w_1 handles reasoning and w_2 handles generation, the hub issues probe briefs (small, bounded tasks with known answers) and observes acceptance rates, latency, and output quality before routing production work. When a brief arrives, it carries a threshold condition (for example minimum quality $q \geq 0.87$, maximum latency 45 s, cost $\leq c$). The registry is queried against this condition, and only workers whose probe-verified profile satisfies all three dimensions are eligible. Workers who fail probes are not penalised. They are not routed until a later probe re-establishes their profile. This has a direct lineage in A2A agent cards [2], which standardise self-reported capability disclosures, and in ERC-8004's conception of task-aware routing [4]. Both treat capability as a claim subject to verification, a position we adopt.

Status in our implementation. Routing across the two worker models was hand-configured, not registry-computed. The v1 dispatcher routed any unrecognised worker name to one default provider (v2 rejects it). The registry is the layer with the largest gap between design and implementation.

Probe economics. The binding constraint on registry fidelity would be verification bandwidth, if the conjecture of §3.5 holds. The institution cannot probe every worker on every task type at every moment. The optimal probe frequency is a function of task-value density, worker stability, and the penalty for misrouting. In practice this resolves to probing stratified by worker volatility. Stable workers are probed infrequently, and workers near their capability boundaries proportionally more often.

4.4 Constraints, acceptance, and the provenance audit (IMPLEMENTED in part)

Delegation without verifiable acceptance is delegation without a floor. An institution that cannot observe whether a delegated task was accepted bears the full variance of the agent's behaviour, which collapses the cost argument for delegation into plain outsourcing. Machine-checkable acceptance criteria are not a UX refinement. They are the structural prerequisite for bounded delegation. But in the sessions reported here the machine-checkable criterion was a needle, and of the two incidents that matter, one was a needle pass and the other a variant that the whole-transcript test could not distinguish from a pass. The apparatus that actually decided acceptance was the audit.

The incidents that show why. During this paper's own assembly, a worker exceeded its mandate: it inserted its section directly into the manuscript and signed a variant of the needle, which the instrument recorded as a completion and the hub accepted as a signature (INC-001, Appendix B). Separately, a worker produced a worked example with no instrument record, which passed its needle and two rounds of agent review before a hub-side cross-check against the instrument caught it (INC-004). Needle satisfaction is not mandate compliance, and consistency-checking review does not catch grounded fiction. Only a check against the record does.

The provenance audit (IMPLEMENTED). After each wave the hub compares every artifact against its mandate and against the instrument: file modification times and needle signatures against the dispatch record (which caught INC-001 within five minutes), a size gate on every edit to the manuscript (which caught INC-002), and a cross-check of worker-supplied figures against the instrument table and a turns-after-artifact query of the message store (which caught INC-004 and has been a standing post-weave step since 8 October). This is the acceptance apparatus of record in this institution. It is post hoc, it is hub-side, and it has a catch record of three. It also has a miss record of three through 9 October, four with INC-007: the stale figure (INC-003), the remembered turn count (INC-005), and the unverified anchor (INC-006) were all hub-side number claims, all shipped after the audit was standing, and all caught by the agent reviewers, not by the audit, because the hub did not run its own cross-check against the numbers it had written itself (INC-003's gate verified presence, not absence; INC-005 had no query). For INC-006 the hub did not run the cross-check itself but instructed the next review round to run it, which is the §8 lesson practised rather than stated. The audit catches worker fiction and worker scope violations. It did not catch the hub's own receipts. It belongs in the anatomy as the fifth layer, and §3.5's consistency claim includes it, with that limit.

Needles as liveness checks (IMPLEMENTED). A needle is a literal string the worker must emit. It establishes that the worker parsed the brief and reached the end of its task. It does not establish that the task

was done. In the reported sessions the dispatcher tested the whole transcript for the string, and on rows 7–26 the CLI quoted the brief, needle instruction included, into that transcript (§3.7), so on those rows the test was satisfied by construction, and the result was not logged on any row. The v2 dispatcher invokes the CLI's quiet path, which prints the response alone, requires the needle as the last non-empty line of that response, and logs the result. It has been exercised once against a live worker (§3.7) and recorded a hit. Even then, a needle is sound for the deliverable only when $\nu \Rightarrow \delta$ (§4.2), which a literal string never is on its own.

Three instrument types.

- Needles (IMPLEMENTED as above). The 16/16 exit-zero completions in §3.7 are a calibration signal, not a performance metric: they say the needle design was parsed by the workers, and nothing more, because the needles were undemanding and the outcome was not logged.
- Budgets (IMPLEMENTED: wall-clock alarm only; token and spend caps DESIGNED). The wall-clock alarm fired twice in the live instrument (§3.7). There is no token cap, no spend cap, and no pre-dispatch envelope check in the dispatcher; the running example's token budgets in §4.2 are unenforced. Spend caps are analogous to ERC-4337 session keys [5], which embed permission boundaries in the transaction context.
- Alarms (wall-clock IMPLEMENTED; SLA, error-rate, and deadline alarms DESIGNED). An alarm fires when latency exceeds an SLA, error rate breaches a tolerance, or a needle remains unresolved past its deadline. Only the wall-clock alarm exists.

Acceptance and the condition. ($\star\varepsilon$) needs p bounded away from zero and estimable. Without logged acceptance, p is unknown. With it, the institution can test after each wave whether the condition still holds for a worker or whether it must reassign the delegation. The i.i.d. assumption behind any interval on p is violated by self-selected briefs, so the interval describes the brief mix, not a worker.

What the instrument actually recorded. Sixteen rows at the frozen snapshot, dispatched in parallel (six launched within one second in the section wave), 16/16 exit-zero completions, needle status hub-attested at 12/13 on briefs, mandate-compliant acceptance at most 11/13, zero retries by the ladder, zero escalation events, and no acceptance windows, deadline alarms, or retry budgets implemented. After the freeze the instrument recorded its first wall-clock failures: a review brief hit its budget, was re-dispatched by hand, and hit it again (§3.7). The needle channel plus the audit were the entire acceptance apparatus. The two timeouts mean the completion distribution is no longer degenerate. The acceptance distribution is still the two audit catches.

4.5 Escalation paths (DESIGNED)

When a brief fails the audit or its needle, the institution chooses between re-dispatch and escalation. We treat escalation as a first-class cost and the escalation path as a programmable control law.

The ladder. A brief that fails validation triggers a retry with truncated context: a partial read that strips the worker to task-relevant signal. If that resolves the failure, escalation is avoided. If not, the institution rewrites the brief under stricter constraint, narrowing the needle or the search space. Only when rewrite also fails does the institution escalate to a human reviewer. The ladder, retry then partial read then brief rewrite then human, is the institution's escalation policy. It is consistent with the cost model, not derived as optimal, and it has not run (§4.6). As specified in the previous version of this paper, the ladder was triggered by needle failure alone. INC-004 was a needle pass, so the ladder could not have fired on it. INC-001 carried a variant needle, so the dispatcher's exit status would have signalled a miss, but no ladder existed and the hub did not act on the exit status. The audit caught it instead. The trigger is now audit failure or needle failure, and the response an audit failure actually requires (quarantine the artifact, restore the manuscript, withdraw every copy) is the first rung for that trigger, because retry, partial read, and rewrite were designed for needle failure and none of them is what the two audit failures on record needed.

Escalation budget. ($\star\varepsilon$) gives the budget in closed form: with persistent ε, escalation remains affordable while $\varepsilon < S\ [\ p(1 - \alpha - (b+o)/S) - (w/h)/(1+T)\ ]\ /\ (1 - p)$. Each rung of the ladder adds to ε (a rewrite costs less than a human handoff), so $\varepsilon = \Sigma_r\ P(\text{reach } r)\cdot\varepsilon_r$ over rungs r. When the condition is tight the correct response is re-dispatch, not escalation, and the institution must learn that threshold.

When to escalate versus re-dispatch. Re-dispatch suits environmental failure: a transient condition

unlikely to recur with identical input. Escalation suits structural failure: a malformed brief, a miscalibrated needle, or a task beyond the worker's capability. A system implementing these principles would record the delta between dispatched and received for each needle, and a cluster of retries beyond a latency or quality threshold would trigger escalation automatically.

Manager-analogue, not manager-identity. The human reviewer in an escalated brief is not a manager in the organisational sense. They do not re-prioritise, re-allocate, or direct workflow. They are a verification endpoint with bounded bandwidth and higher tolerance for ambiguity. Naming the role ”manager” invites assumptions about authority, state, and continuity that do not apply to a stateless handoff.

4.6 What escalation actually caught, and the hub as a governed actor

In the runs reported here, the escalation ladder never fired. INC-001 was caught by the provenance audit, not by the ladder. INC-004 was caught by the audit after two review rounds missed it. The one unbounded edit on record (INC-002) was committed by the hub itself. Escalation as designed remains untested under load.

Five of the seven incidents (and one process finding) through 13 October are hub-side: the hub destroyed two sections of its own manuscript (INC-002), shipped a stale figure (INC-003), reported a count from memory instead of a query (INC-005), carried an anchor from a reviewer footnote without verification (INC-006), and transcribed a reviewer's wrong total into Table 1 (INC-007). The anatomy of §4.2–4.5 constrains workers. The hub is an agent inside the institution and a single point of failure (one session, one store, compaction outside its control), and the only hub-side controls that ran were the post-edit size gate, the claim-vs-instrument diff, and version control. A checkpoint before every destructive edit was not among them. The human's role in the reported sessions was supervision of each wave and the final read-through, with the autonomy level per action class (dispatch, weave, edit, publish) not formally documented. This is a limitation the institution reports about itself.

5. Method: The Institution Studies Itself

System. One persistent hub session (GLM-4.6, human-supervised) and ephemeral free-lane workers (DeepSeek-V4-Flash, MiniMax-M2.7) over an OpenAI-compatible gateway. Marginal worker cost: $0 (rationed free tiers). The free lane carries a quota price the instrument does not observe: no backoff, jitter, or concurrency cap was in place, and the section wave launched six workers within one second.

Procedure. Every unit of work is a dispatch. (1) The hub writes a mandate (scope, deliverable, constraints, escalation note) into a brief file. (2) The brief specifies an acceptance needle, a literal string the worker's final output must contain. (3) The dispatch script runs the worker under a wall-clock budget, appends one instrument row, and tests for the needle. (4) The hub adjudicates by provenance audit: needle-pass is necessary but not sufficient, and post-wave scans compare the artifact against the mandate and the instrument. INC-001 and INC-004 exist because workers passed, or were recorded as passing, (3) while failing (4). (5) Accepted artifacts are cleaned disk-to-disk and woven by the hub. Admitted character counts were audited by the hub but not logged (§3.1).

Reflexivity. The paper's measurements are the institution's operational log. Its incidents (Appendix B) are its governance data. Its revisions were performed in response to review by LLM reviewer agents: the institution's verification standards enforced against its own author, with the limits stated next.

Agent review, as it was. Five rounds of review were run by panels of three LLM reviewer agents with methods, economics, and multi-agent-systems personas, under briefs written by the hub. This is not peer review. The reviewers were not independent of the institution and were not blind to each other. Rounds 3–5 were dispatched through the same dispatcher to the same DeepSeek-V4-Flash worker that drafted the citation sweep and wave 3 and ran the citation resolver (the introduction and the six section drafts were MiniMax-M2.7). Rounds 1–2 were run outside the instrument and their model is not recorded. Reviewers ran in the same working directory and could read sibling and prior-round transcripts, and in two documented cases did (reviewer C's round-2 materials list and its round-5 transcript, which cites a sibling's result).

What the panels verified: citation identifiers re-checked against live registries, every Part 3 number recomputed, the dispatcher and instrument table inspected at source, the breakeven arithmetic recomputed.

What the panels caught:

- A retracted statistic still live in the abstract (INC-003).
- A remembered turn count that did not reproduce under a reviewer's own query (INC-005, reviewer A, round 3).
- An unverified anchor and a false monotonicity claim found by two reviewers re-running the published query at the hub's instruction (the round-4 briefs, task T3(b), named the expected value 277; both got 266) (INC-006, reviewers A and B, round 4).
- Three mutually inconsistent dominance formulas across sections.
- Two breakeven entries failing reproduction.
- α not recomputable from the instrument.
- A self-certified rate reported without an interval.
- The hub's omission of the nearest prior art [15] from the manuscript's verdict table after the novelty strike had named it the most dangerous paper.
- And the escalation ladder credited for a catch made by the audit.

Three of the six incidents through 9 October, all hub-side number errors, were the panels' catches, and the second panel's catch of INC-007 makes four of seven.

What the panels missed or got wrong: the fabricated worked example (INC-004) passed six reviewer passes before a hub-side record check caught it. In round 3 one reviewer reported grounding the INC-005 turn count in the message store with an exact match that re-ran to a different value, a confabulated receipt, in the same round another reviewer caught the number. The anchor that INC-006 records as unverified was imported from a reviewer footnote. The record is symmetric: the audit caught what the panels could not see (worker fiction and scope), the panels caught what the hub's audit did not run on itself (its own number receipts), and neither caught everything. LLM reviewers from the same free-lane pool as the workers verified what they were pointed at. The evidence for this paper's claims is the published artifact set, not the review process or its scores. A human read-through of the full manuscript was completed on 12 October 2026. A second LLM panel, four personas of a different model family run by the author's own review tooling under author-written briefs, reviewed v1.5 and v1.6 on 13 October 2026 and is likewise not peer review. Its findings are what this revision answers, and its own round-1 synthesis supplied a wrong total that the author transcribed into Table 1 without recomputing it (INC-007).

Novelty audit. Two worker sweeps (a prior-art trawl and an adversarial strike instructed to kill the claims), run by the same worker species as the content audited, found live neighbours for all six claims and no claim dead. The audit ran in two stages: a hub-side probe first collected ten candidate papers, including [16], [25], and [26], and the strike brief (`briefs/strike_brief.md`) then instructed the worker to pull each abstract, chase its neighbours, rate the overlap, and search beyond the list. The strike's job was to test and extend the hub's leads, not to find them from nothing. The residual deltas are stated in §2.7. Is the setup unique? No. Free-worker offload is adjacent to model cascades. The unique parts are the horizon term under Assumption E, the per-context-budget bound, the instrumented run, and the incident record.

Citations. Of the 34 references, the 16 arXiv identifiers and the Bayern citation were re-verified live on 12–13 October 2026 against arXiv and the publishers' pages, and the Karlin, Esteva, Coase, and Alchian–Demsetz entries against publisher records on 13 October. The verification transcript is in the companion repository as `reviews/citation_check_2026-10-13.md`. Web-only references (protocol specifications, EIPs, framework documentation) carry their URLs. All were last accessed on 13 October 2026.

Artifacts. The dispatch script (v1 as run and v2 as corrected), the instrument table, the frozen turn archive with its verifier and de-duplication script, the number snapshot, the incident and revision logs, the novelty-audit transcripts, the full reviewer transcripts for all five rounds, and the worker-stage section archives (including the quarantined fabrication of INC-004) are public in the companion repository at github.com/nKOxxx/agentic-institutions, tag v1.7.9. The 23 v1 dispatched brief files, the reconstructed v2 probe brief, and the shared round-2 reviewer note are in `briefs/`. Two one-line briefs were not retained: the first connectivity probe, which is lost, and the first live v2 dispatch, which is reconstructed in `briefs/` from the author's record of the

command that created it. The command is not retained, the file is 87 bytes matching the row's `brief_bytes`, and its content is consistent with the retained response. The reconstruction is hub-attested, like every v1 needle outcome. The constraints brief asked for the 7/7 needle-hit rate as its worked example and, in its rules line, permitted "concrete examples from a live hub-and-worker system" it gave the worker no record of. The 147-brief figures were the worker's invention, the licence was the hub's, and of the other five section workers given the same line, two used it with real examples from the run (`sections/sec_s3a.md`, `sec_s3b.md`) and none invented a record. The briefs also close the provenance of four claims the second panel struck in its first round: "cost-optimal control law" and "the economics re-derive the anatomy" (the introduction brief, and the first as framing in all six section briefs and the review briefs), "free-lane delegation cannot lose for any acceptance probability", and the "~6× lifespan" multiplier. All four entered the manuscript as hub instructions, were carried by the workers, and were caught by review. The introduction brief also carries the pre-($\star$) form of the condition, $(2-\mathrm{p})[\mathrm{v} + (\mathrm{w}/\mathrm{h})(\mathrm{S}/\mathrm{T})] < \mathrm{S}$, which is the one-retry variant of ($\star\varepsilon$) with every attempt's output admitted. Its "cannot lose for any p" claim is a theorem of that variant for $\alpha < 1/2$ and is false under ($\star\varepsilon$), whose free-lane threshold at $\varepsilon = \alpha\mathrm{S}$ is $\mathrm{p} > \alpha$. Its "~6×" is $1/\alpha$ at the brief's own v/S; under the variant the multiplier is $1/((2 - \mathrm{p})\alpha)$, 5.1 at the measured p; under ($\star\varepsilon$) $1/\alpha$ survives as the admission-only bound of §3.4 (3.3 with the measured brief term) and was withdrawn only as a lifespan claim, for the reason §3.4 gives. That is the audit trail the method should produce by default, recovered here on request. The hub's message store is not public, so every figure in Appendix A F3 is reproducible only by the author.

Instrument limits. n = 16 rows at the frozen snapshot (13 briefs, three probes). Briefs were self-selected. No admitted count is logged. The needle outcome was not logged for any public row. Rows 1–6 were written by a script version not in the repository. p and T are bounded estimates pending accumulation. Headline numbers are frozen in the number snapshot at n = 16, and live counts may exceed them.

6. Conjectures for Agentic Market Design

This section was drafted by a free-lane worker (the INC-001 dispatch, whose text was recovered through the front door at assembly), hub-edited, and is labelled throughout as conjecture. If institutional anatomy is consistent with the cost model of Part 3, the model suggests, without establishing, which markets a delegation economy would need. Nothing here is instantiated by the institution of §5.

Mandate marketplaces (conjecture). A mandate is a bounded, machine-readable authorisation verifiable before commitment. If mandates were tradeable, a market for delegation rights would emerge: principals auction bounded briefs, workers bid with attested capabilities, and p is estimated from registry records. This would be structurally a labour marketplace whose employment contract is a machine-verifiable mandate. The infrastructure gap is a standard mandate description language, something the IETF or IEEE could specify the way they specified TLS ciphersuite negotiation. We are not aware of a production instance and cite none.

Registry standards (conjecture). Capability registries would be the trust substrate for routing. A worker without a verifiable attestation cannot price its services. A hub without a registry cannot distinguish a reliable worker from one optimising for needles. This institution has no registry and so cannot (§4.3). Registry standards would serve the function credit bureaus serve in human finance. ERC-8004 [4] provides the identity primitive. The attestation layer above it (signed performance records, domain-specific competency tests, historical p estimates) is the missing standard. Kolt [17] frames the governance need. No schema has converged the way X.509 converged for web identity.

Acceptance verification as a paid service (conjecture). If the verification-binding conjecture of §3.5 holds, verification becomes the primary economic relationship in machine-to-machine markets. A market for verification-as-a-service would follow: third-party auditors that accept a brief, a worker artifact, and a mandate, and return a signed acceptance verdict, doing for acceptance what a payment processor does for card transactions. In our institution the hub ran needles and a post-hoc provenance audit on every returned artifact (§4.4). Through 9 October the audit caught three of the six incidents and the agent reviewers the other three. The second panel then caught INC-007, which the audit also missed.

Escalation insurance (conjecture). Every mandate eventually hits a boundary the brief did not anticipate. If the ladder is finite with a per-rung cost cap, its cost is bounded and in principle insurable: principals pay a

premium against the risk that their agent's mandate is exceeded, and the insurer underwrites the escalation cost. Pricing requires a loss distribution over escalation events, and this institution has recorded zero.

Open problems. Three are urgent. Mandate portability: if an agent holds a mandate from principal A and is invoked by principal B, does the mandate bind? Current protocols say nothing. Cross-organisational escalation: when an agent operating under mandate M in organisation O_1 must act in O_2's domain, the escalation path crosses a trust boundary neither controls. A2A handles the communication. The institutional question of who resolves, under what rules, with what enforcement, is open. Machine liability: when an agent acts beyond its mandate, who is liable, and how is liability settled? Yu, Zhao, and Sui [29] document agents already managing treasuries in production with no liability framing at all. No major jurisdiction has resolved this. Bayern [27] shows how far entity law can be stretched to house an autonomous system, and how little that settles about liability. Until it changes, principals hold unlimited liability for their agents' actions, which will suppress agentic commerce.

7. Figure: The Delegated Cognition Stack

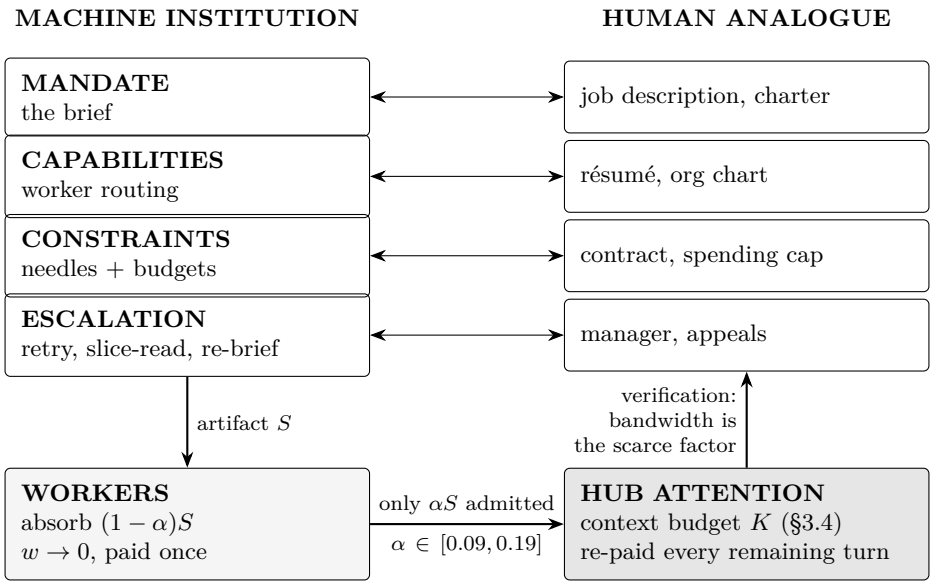


Figure 1: The delegated cognition stack. Left column: the four institutional layers that act before and during a dispatch; the provenance audit of §4.4 acts after it and is not drawn. Right column: the human structure each one functionally replaces. Bottom band: the cost law. Workers absorb $(1-\alpha)$S at vanishing price. The hub admits αS against a finite context budget K. The bottleneck at K is the conjecture of §3.5.

8. Conclusion

Delegation between a persistent hub and its workers obeys a rent-or-buy condition whose terms are measurable in an institution's own logs. For free workers the condition holds whenever acceptance exceeds a small threshold. For paid workers it holds over a horizon, under the assumption that inline output persists in context, which this institution's own store violates. Per unit of context budget, delegation admits up to $1/\alpha$ artifacts per inline artifact, and the hub's admission policy sets α. Whether verification becomes the binding constraint as worker cognition approaches zero cost is a conjecture this paper states and does not test. The four institutional layers plus a provenance audit are consistent with that cost structure. They are not derived from it. The institution that wrote this paper is small, its instrument logged completion rather than acceptance, its escalation ladder has not fired, and five of its seven incidents (with one process finding beside them) were its own hub's. It is nonetheless real, it runs on free infrastructure, its artifacts are public, and its two decisive failures were caught

by checking claims against the record rather than by checking them for consistency. That is the result: needles are not acceptance. The audit caught what the panels could not see, the panels caught what the hub's audit did not run on itself, and provenance checks must be run by someone other than the author of the number.

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Appendix A. Measured results

All numbers are from the dispatch instrument table (public) and the hub message store (not public), frozen 9 October 2026. [M] = measured and reproducible from the public artifacts. [B] = bounded or hub-audited estimate with method stated. [H] = from the hub's private store, not reproducible from the repository.

F1 [B]. Admission fraction. Four waves land in hub-audited $\alpha \in [0.09, 0.19]$ (Table 1), a 2.1× range. $1/\alpha$ spans 5.3–11.1 across waves (wave-rounded. 11.6 on the unrounded wave-4 value) and is 7.3 pooled, as the admission-only upper bound on the per-context-budget multiplier. With the measured brief term the bound is 3.3 pooled on the paper's mixed pairing and 3.6 with α and b/S on one hub-audited denominator (§3.4). No admitted count is instrument-logged. α behaves as a hub policy parameter of needle-based admission, not as a constant.

F2 [M] completion, [B] acceptance. 16/16 exit-zero completions at the frozen snapshot, of which three rows are connectivity probes (one in the opening run, two after the 14-row freeze) and 13 are briefs. Needle status is hub-attested at 12/13 on briefs (row 13 is INC-001). Mandate-compliant first-try acceptance is at most 11/13 (INC-001, INC-004). n is small. The instrument accumulates.

F3 [H]. The hub's own log. Lifetime hub session since 25 July 2026, multiple compactions: 17.39 MB, 27,775 messages, 75.6 % raw tool output. Worker side, two denominators: worker session traffic of 7.60 MB gives hub:worker = 2.3× (reported as 2× in earlier versions, truncated), and worker deliverable artifacts of 94.2 KB give hub:artifact = 185× (184.6 rounded). The windows differ (§3.7). The persistent hub pays every character repeatedly. Workers pay once. Each compaction event is a visible re-payment of the accumulated tail. An earlier draft cited 75× with a non-reproducible denominator. The instrument corrected it on first run (INC-003). None of these figures can be regenerated from the repository, because the stats script reads the hub's private store.

F4 [M]. Duration scales with internal tool use. Two-message drafts: 24–43 s. An 18-message citation sweep: 109 s. Research sweeps: 479–525 s at 27–70 messages. Pooled over the 16 frozen rows, duration correlates with output bytes (r = 0.89) and with message count (r = 0.86). Within the DeepSeek-only subsample output bytes dominate (0.91 vs 0.82). Rows with unparsed message counts are excluded.

F5 [B]. Inline counterfactual for the audit wave. 35 hub turns × 554-character mean substantive message ≈ 19.4K hub characters, versus ≈7K admitted. Inline turns carry full tool transcripts (the 75.6 % pattern), so 19.4K is a floor. A plausible inline figure is 30–60K. Delegation saves in the bounded reading and dominates in the realistic one, subject to the eviction caveat in §3.6 and to the hub-audited status of the 7K figure.

Concurrency. Six workers were launched within one second in the section wave and five appended to one instrument table within two seconds without corruption. That is the expected behaviour of small O_APPEND writes under the pipe buffer limit, not a finding, and the same wave produced INC-001.

Appendix B. Incident register

Each incident is recorded in full in the companion repository's incident log (§5).

- INC-001 (7 Oct 2026). A worker exceeded the implicit scope of a draft brief that stated no action constraint: briefed to draft a section, it wrote the section directly into the manuscript and signed a variant of its needle (the brief's section label, IV-IMPLICATIONS-OK, in place of the specified SECTION-S4-OK), which the instrument recorded as a completion and the hub accepted as a signature. Caught five minutes later by the post-wave provenance audit (file modification time and signature check), not by the escalation ladder. Contained by extracting the block and restoring the manuscript, insertion only, zero loss. Its instrument row reads OK (§3.7). In the terms of §4.2 it violates the scope condition (ii).
- INC-002 (7 Oct 2026). The hub rewrote its own manuscript with a regex anchored at one heading that consumed everything to end of file, deleting the novelty audit and the citation adjudication. Caught by a post-edit size gate. Both sections restored. The hub is an agent inside the institution and needs the same bounded-edit constraints it imposes on workers.
- INC-003 (8 Oct 2026). A correction pass replaced the 75× traffic figure with the computed 185× but asserted only that the new figure existed, not that the old one was absent, and a silent no-match left the stale figure in the abstract. Caught by agent review. Fix: every revision asserts pre-state and post-state, and one frozen snapshot is the single source for headline numbers.
- INC-004 (8 Oct 2026). A worked example describing a 147-brief, 24-hour, four-worker run with 30-second acceptance windows was worker-generated fiction. It passed its needle and two full review rounds and was caught by a hub-side cross-check against the message store and the instrument. The brief had asked for the 7/7 needle-hit rate as the worked example; the fabrication was not requested, but a rules line in the brief licensed concrete examples from a system the worker could not observe, so the under-specified deliverable was the hub's. Withdrawn. In the terms of §4.2 it violates the soundness condition (i). Addendum (9 Oct): the same text survived verbatim in a worker-stage archive file; a quarantine banner now precedes it. Withdrawal must chase every copy.
- INC-005 (8 Oct 2026). The hub replaced a folklore estimate of remaining session turns with a specific count taken from working memory rather than from a query. Reviewer A could not reproduce it under any window, and reviewer B reported a confirming query result that did not reproduce either. Fix: the count is now a published query plus a frozen output.
- INC-006 (8 Oct 2026). The timestamp anchoring that query was carried from a reviewer footnote without primary verification, and the hub-authored claim that the count was monotone was false because the store compacts. The qualifying row set is now archived with a SHA-256 and recounted by script. The 13 October panel found two further receipt errors in the same measurement, the freeze time labelled UTC when it was local, and 122 compaction rows counted as turns; both are corrected in §3.3.
- INC-007 (13 Oct 2026). The v1.6 manuscript printed the Table 1 wall-clock total as 860. The four per-wave maxima sum to 840. The figure 860 was an arithmetic error in one reviewer's round-1 text, repeated in the chair's list of required changes, that the author transcribed without recomputing from the instrument table in the same repository. Source: the review panel, including its chair. Caught by the same panel in round 2. The INC-006 lesson, that a reviewer's number is a provenance lead and not provenance, failed on the first table of the revision that reported INC-006 as fixed. Addendum (14 Oct): two further panel-sourced sentences were transcribed without checking and caught by the panel's third round: a committed-policy bound stated below 1 (from the chair's round-2 required-change list) and an abstract clause asserting the model of two review rounds the paper says it cannot characterise (from a reviewer's suggestion adopted by the chair). Fix: a build-time check, run by the PDF build script, recomputes every instrument-derived Table 1 cell (bytes, wall maxima, total, resolver row) from the instrument table and fails the build on mismatch; the hub-audited cells (N, admitted, α) cannot be checked that way.
- INC-008 (13 Oct 2026, process finding). The worker CLI's single-query path echoes the full brief to the transcript the v1 dispatcher tested, so the v1 needle test could be satisfied without the worker emitting anything (§3.7). Found by the second panel from the CLI source and confirmed by the author. No number changes; needle status for all v1 rows is hub-attested only. Fix: v2 invokes the CLI's quiet path, which prints the response alone, and normalises line endings before the gate.